HealthScope: A Public Health Analytics and Diabetes Risk Prediction Platform

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ARTICLE INFORMATION AND ABSTRACT

ARTICLE INFORMATION

Keywords: Public health analytics; Diabetes indicators; Logistic Regression; Exploratory data analysis; Streamlit; Health risk classification

ABSTRACT

Health surveys contain information about people's health conditions, daily habits and access to care. Exploring these records manually can make it difficult to identify patterns across many indicators. This paper presents HealthScope, a web-based application that combines public-health data analysis with machine-learning-based diabetes-status classification. The system uses 253,680 records, 21 input features and a three-class target: No Diabetes, Prediabetes and Diabetes. A Logistic Regression model is integrated into a Streamlit dashboard that provides a health-information form, prediction results, population-level charts and model evaluation. The reported full-dataset evaluation achieved 65.41% accuracy, 84.89% weighted precision, 65.41% weighted recall and 72.54% weighted F1-score. The results show that the model performs unevenly across classes, particularly for Prediabetes. Since evaluation was carried out on the complete dataset, including training records, the metrics do not establish performance on unseen individuals. HealthScope is an academic analytics prototype and is not intended for medical diagnosis.

The main idea behind HealthScope is to make a large health-indicator dataset easier to explore and to demonstrate how a classification model can use several inputs together. The application presents both the predicted category and the data patterns that provide context for interpreting the result.

1. INTRODUCTION

Diabetes is a long-term health condition whose occurrence is associated with several demographic, lifestyle and health-related factors. Public-health surveys collect information such as age, body mass index, blood pressure, cholesterol, physical activity, smoking habits and general health. When these details are available in large datasets, they can be analysed to understand how different indicators appear across population groups.

Reviewing thousands of survey records one by one is not practical. Data analysis tools help summarize the information, while machine-learning methods can learn patterns from labelled examples and apply them to new input records. However, a useful system should do more than return a class label. It should also make the dataset and model behaviour easier to understand.

HealthScope was developed with this goal. It combines data loading, preprocessing, exploratory data analysis, diabetes-status classification and visualization in one browser-based application. The workflow follows five connected stages: Collect, Clean, Analyze, Predict and Visualize. Users can enter health and lifestyle information through a form, receive a predicted category and explore charts that summarize the dataset.

The three target categories are No Diabetes, Prediabetes and Diabetes. This distinction is important because the three groups are not equally represented in the source data, and the model does not perform equally well for each group. HealthScope therefore includes model-performance information alongside the prediction interface.

The project is intended as an academic demonstration of a complete data-science workflow. It does not replace a clinician, laboratory testing or professional medical advice. A predicted class should be treated as a model output based on survey patterns, not as a diagnosis.

1.1 OBJECTIVES

- To organize a large public-health survey dataset for analysis.
- To explore the relationship between diabetes status and selected health or lifestyle indicators.
- To build a three-class classification model using Logistic Regression.
- To provide an easy-to-use Streamlit interface for entering health-related information.
- To display dataset summaries and model evaluation results so users can interpret the output more carefully.

1.2 CONTRIBUTIONS

The contribution of the project is mainly its integration of several functions in one interface. The application connects an input form with a saved preprocessing scaler and trained classifier, while the analytics section presents population-level patterns. The performance section also makes class-wise weaknesses visible instead of relying only on a single accuracy value.



Fig. 1. High-level HealthScope workflow shown in the project documentation.

2. RELATED WORK

Public-health data analysis draws on survey systems that record health conditions, behaviours and demographic information. The Behavioral Risk Factor Surveillance System (BRFSS) is one such source and provides survey information used in population-health research [1]. The Diabetes Health Indicators dataset used in HealthScope is derived from BRFSS information and makes it possible to examine diabetes status alongside a range of health indicators [2].

In machine learning, classification methods are used when the required output belongs to a set of categories. Logistic Regression estimates the probability of an input belonging to each class. Although its name includes the word regression, it is widely used for classification tasks. Its relatively straightforward structure makes it a useful baseline for projects in which the relationship between input variables and predicted categories needs to be examined [3].

Data preparation is an important part of such a workflow. Features may have different numerical ranges, so standardization can transform them to a comparable scale. Scikit-learn provides StandardScaler, which learns the mean and standard deviation from the data used to fit it and applies the same transformation to later inputs [4]. A saved scaler is therefore used by HealthScope at prediction time to keep user-entered data consistent with the model's training representation.

Interactive dashboards are also useful in applied data-science projects because they bring inputs, outputs and visual summaries together. Streamlit provides a Python-based way to create web interfaces with forms, metrics, tabs and charts [5]. HealthScope uses this approach to make the prediction workflow accessible without requiring users to run model code directly.

The project combines these established components in a single public-health analytics prototype. The focus is not to claim a new classification algorithm, but to demonstrate a practical pipeline in which survey data, preprocessing, classification and visual interpretation are connected. The current implementation uses Logistic Regression; comparison with Decision Tree, Random Forest and K-Nearest Neighbours is treated as future work.

2.1 SUMMARY OF TECHNOLOGIES AND THEIR ROLE

COMPONENT — Role in HealthScope | Reason for use

- BRFSS-derived health indicators: Provides survey records and target labels. Contains demographic, lifestyle and health information
- Pandas and NumPy: Load and handle tabular/numerical data. Support data inspection and transformation
- StandardScaler: Standardizes the 21 model inputs. Keeps input scale consistent with training
- Logistic Regression: Predicts one of three diabetes-status classes. Provides a baseline multi-class classifier
- Streamlit: Builds the interactive web dashboard. Connects form inputs, metrics and visualizations
- Matplotlib: Displays model evaluation visuals. Supports confusion-matrix presentation
- Joblib: Stores and reloads model and scaler. Avoids retraining every time the app starts

2.2 RESEARCH GAP AND PROJECT MOTIVATION

The motivation for HealthScope is the need for a compact workflow that does not separate model prediction from data exploration. A model output on its own gives limited context, particularly when a dataset is imbalanced. By including population summaries, correlation charts and class-wise evaluation, the application helps users see both the patterns in the data and the limits of the classifier. The project remains a prototype and requires stronger validation before it can support real-world decisions.

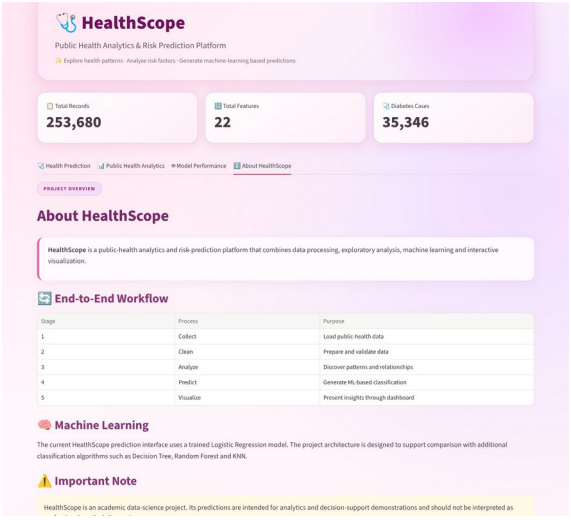


Fig. 2. HealthScope dashboard overview with dataset summaries and explanatory sections.

3. SYSTEM MODEL AND PROBLEM STATEMENT

HealthScope takes a set of health-related attributes as input and produces a diabetes-status category as output. The input consists of 21 features that describe clinical indicators, lifestyle habits and demographic or personal information. The target variable is Diabetes_012, which contains three labels: 0 for No Diabetes, 1 for Prediabetes and 2 for Diabetes.

The problem is a supervised multi-class classification task. During model training, the classifier learns relationships between the input features and the known target labels. During prediction, a user enters values through the dashboard and the application prepares them in the same format as the training data. The model then returns one of the three categories.

3.1 INPUT FEATURES

FEATURE GROUP — Examples included in the dataset

- Clinical indicators: High blood pressure, high cholesterol, cholesterol check, BMI, stroke history, difficulty walking, general health
- Lifestyle indicators: Smoking, physical activity, fruit consumption, vegetable consumption, alcohol-related indicator
- Healthcare access: Health-care coverage and cost-related access indicator
- Personal and demographic: Age group, education, income, sex, and other demographic attributes
- Health history: Physically unhealthy days and mentally unhealthy days

The model uses the exact 21 feature columns in the source dataset. In the user interface, age is entered in years and then converted into the age-group coding expected by the dataset. This conversion is necessary because the form's human-readable input and the model's numerical representation are not identical.

3.2 TARGET CLASSES AND CLASS IMBALANCE

The dataset contains 253,680 records. No Diabetes is the largest class with 213,703 records (84.24%). Diabetes contains 35,346 records (13.93%), while Prediabetes contains 4,631 records (1.83%). This uneven distribution is a central characteristic of the task. A model may perform well on the majority class while missing a substantial share of a smaller class.

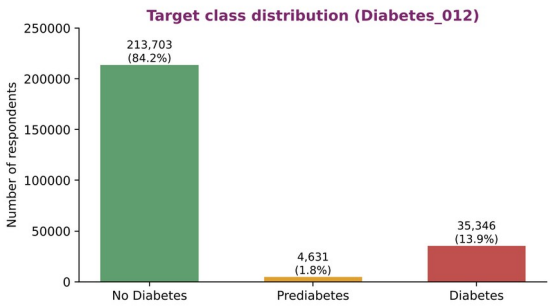


Fig. 3. Target-class distribution in the HealthScope dataset.

3.3 PROBLEM CHALLENGES

- Class imbalance: Prediabetes has far fewer records than the other two classes.
- Different feature scales: health indicators include binary, ordinal and numerical values.
- Consistent input representation: user-entered values must match the training feature order and encoding.
- Interpretation: the predicted class should not be confused with a medical diagnosis.
- Evaluation: full-dataset metrics do not show how the model will perform on unseen records.

3.4 DESIGN OBJECTIVES

The system is designed to accept a user profile, apply the saved preprocessing transformation, generate a three-class prediction and display a clear result. In parallel, the dashboard summarizes the dataset and presents model metrics. The design also aims to keep the prediction process simple enough for a user who does not need to interact with the Python implementation.

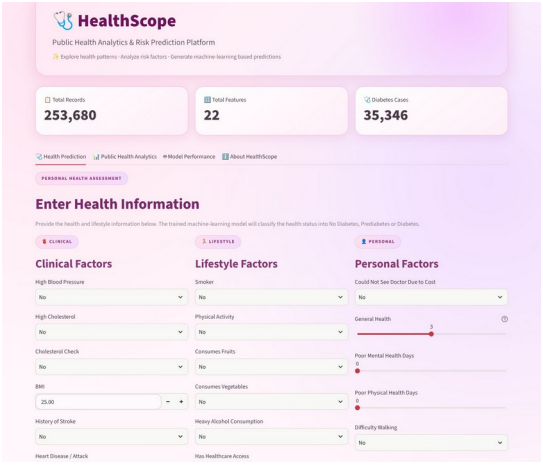


Fig. 4. Health information input form used to collect the model features.

4. PROPOSED METHODOLOGY

The HealthScope methodology connects data preparation, model training, user-input processing, prediction and visualization. The same feature definitions and preprocessing transformation are used throughout the workflow so that a prediction made through the web form is compatible with the trained classifier.

4.1 DATASET LOADING AND INSPECTION

The application loads the CSV file healthscope_data.csv, which contains 253,680 rows and 22 columns. One column is the target Diabetes_012 and the remaining 21 columns are input features. Initial inspection is used to understand the shape of the dataset, column types, class distribution and data quality. The project documentation reports no missing values and 23,899 duplicate rows (9.42%). Duplicate rows were retained in the data used by the dashboard.

4.2 DATA PREPROCESSING

Preprocessing prepares the raw input so that it can be passed to the classifier. The application maintains the feature names and their expected order. This is important because a model trained with a fixed feature arrangement may interpret a value incorrectly if the columns are rearranged.

- Read the dataset and separate the predictor columns from the Diabetes_012 target.
- Use the expected data representation for the model features.
- Convert the age entered by a user into the corresponding age-group value from 1 to 13.
- Create a one-row DataFrame with the same feature names and order used during training.
- Apply the saved StandardScaler to the input row.

StandardScaler uses the mean and standard deviation learned during fitting. The same fitted scaler must be reused for later inputs; fitting a new scaler on a single user record would produce a different transformation and would not match the model's training setup [4].

4.3 LOGISTIC REGRESSION MODEL

HealthScope uses a multi-class Logistic Regression classifier from Scikit-learn. The saved configuration uses L2 regularization, C = 1.0, the lbfgs solver, a maximum of 1,000 iterations and random_state = 42. The project documentation reports that the model converged in 11 iterations. The model receives the standardized 21-feature input and returns a class prediction.

Logistic Regression estimates a score for each possible class and converts these scores into class probabilities. The predicted label corresponds to the class selected by the model. In HealthScope, the output is mapped to a readable label: No Diabetes, Prediabetes or Diabetes. The use of one classifier keeps the current system relatively simple, although it also means that the project has not yet established whether another algorithm would perform better.

4.4 PREDICTION WORKFLOW



Fig. 5. HealthScope's end-to-end workflow from data handling to prediction and visualization.

At application startup, the trained model and scaler are loaded using Joblib. When a user submits the form, the application collects the input values, converts the age field, arranges the features, standardizes the row and calls the model's prediction function. The returned class is then displayed in the result section. The dashboard also provides an educational disclaimer so that users understand the intended scope of the output.

4.5 MODEL EVALUATION MEASURES

The project reports accuracy, precision, recall and F1-score. Accuracy represents the proportion of all records classified correctly. Precision indicates how many records assigned to a class actually belong to that class. Recall indicates how many actual records of a class were identified. F1-score combines precision and recall using their harmonic mean. Weighted averages account for the number of records in each class, while macro averages give each class equal weight.

A confusion matrix gives a more detailed view by comparing actual classes with predicted classes. It helps show which categories are being confused and is particularly useful when class frequencies differ substantially.

5. SYSTEM ARCHITECTURE AND IMPLEMENTATION

HealthScope is implemented as a Python-based web application. Streamlit provides the interface, while the supporting Python code handles data access, preprocessing, model loading, prediction and evaluation. The system is organized into connected layers rather than requiring users to run each data-science step separately.

5.1 MAIN SYSTEM COMPONENTS

LAYER / COMPONENT — Function

- Data layer: Loads the health indicator CSV file and makes the dataset available for analysis.
- Preprocessing layer: Prepares the 21 feature values and applies the saved standardization transformation.
- Machine-learning layer: Loads the trained Logistic Regression model and generates class predictions.
- Prediction interface: Collects user inputs and displays the predicted diabetes-status category.
- Analytics layer: Presents class distribution, health-indicator charts and summary statistics.
- Performance layer: Displays evaluation metrics and the confusion matrix.

The model and scaler are loaded when the Streamlit application starts. This avoids requiring the model to be trained again every time a user opens the dashboard. Joblib is used to save and reload these objects. The current implementation runs the Streamlit interface and Python logic in the same application process; it does not use a separate REST API backend.

5.2 USER INTERFACE

The dashboard is divided into four main sections. The Health Prediction section contains input fields grouped around clinical, lifestyle and personal or demographic information. The Public Health Analytics section displays dataset-level summaries and charts. The Model Performance section presents evaluation metrics and the confusion matrix. The About HealthScope section explains the project and its workflow.

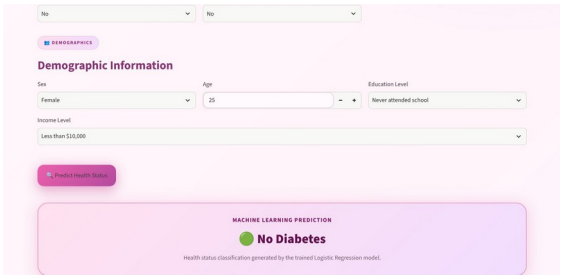


Fig. 6. Example prediction output shown by the HealthScope interface.

5.3 TOOLS AND LIBRARIES

- Python is used for the application logic and machine-learning workflow.
- Streamlit is used to build the interactive dashboard and input forms.
- Pandas and NumPy are used for tabular data handling and numerical operations.
- Scikit-learn provides Logistic Regression, StandardScaler and evaluation metrics.
- Matplotlib is used for plotting the confusion matrix and other analytical charts.
- Joblib is used to save and load the trained model and scaler.

5.4 RUNNING THE APPLICATION

The application is intended for local execution. After installing Python and the project dependencies, the dashboard can be started from the project directory with the command shown below.

python -m streamlit run app.py

After the command runs, Streamlit opens the dashboard in a browser. Users can then access the prediction, analytics, performance and information sections.

5.5 INTERFACE AND ANALYTICS OUTPUTS

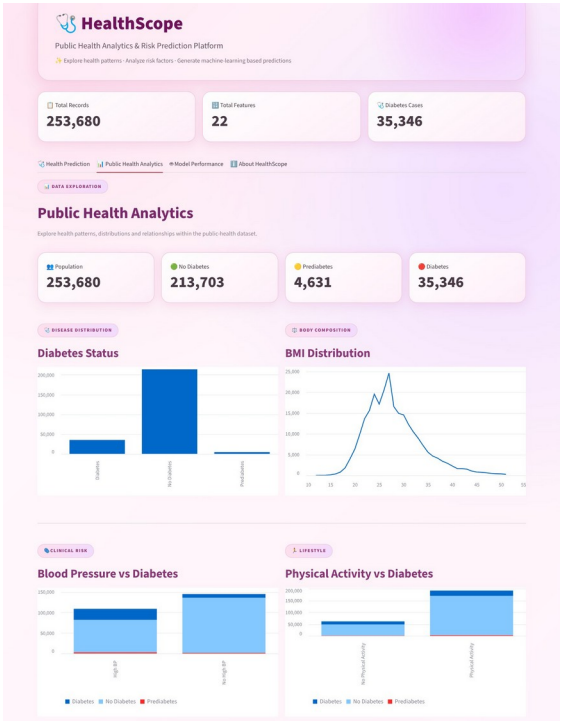


Fig. 7. Examples of health analytics charts displayed in the dashboard.

6. EXPERIMENTAL RESULTS AND DISCUSSION

The results presented here are taken from the HealthScope project documentation. The reported evaluation uses all 253,680 records in the dataset. The overall metrics are 65.41% accuracy, 84.89% weighted precision, 65.41% weighted recall and 72.54% weighted F1-score. The macro-average F1-score is 42.84%.

METRIC — Reported value | Interpretation

- Accuracy (65.41%): Share of all records classified correctly.
- Weighted precision (84.89%): Precision averaged with class support as weights.
- Weighted recall (65.41%): Recall averaged with class support as weights.
- Weighted F1-score (72.54%): Weighted balance of precision and recall.
- Macro F1-score (42.84%): Unweighted average of the class F1-scores.

6.1 CLASS-WISE PERFORMANCE

The class-wise values reveal that the overall metrics hide important differences. The model has its strongest precision for No Diabetes, the majority class. Performance is substantially weaker for Prediabetes, which has the smallest number of records. The reported class-wise values are summarized in Table 2.

TARGET CLASS — Precision | Recall | F1-score | Support

- No Diabetes — Precision: 94.89%, Recall: 67.21%, F1-score: 78.69%, Support: 213,703
- Prediabetes — Precision: 3.16%, Recall: 29.19%, F1-score: 5.71%, Support: 4,631
- Diabetes — Precision: 35.15%, Recall: 59.22%, F1-score: 44.11%, Support: 35,346

The model identifies 20,931 of the 35,346 Diabetes records, corresponding to 59.22% recall. For Prediabetes, it identifies 1,352 of 4,631 records, corresponding to 29.19% recall. Its precision for Prediabetes is only 3.16%, which means that many records predicted as Prediabetes belong to another class. This result shows why the performance of a classifier should be examined for each class, not only through accuracy.

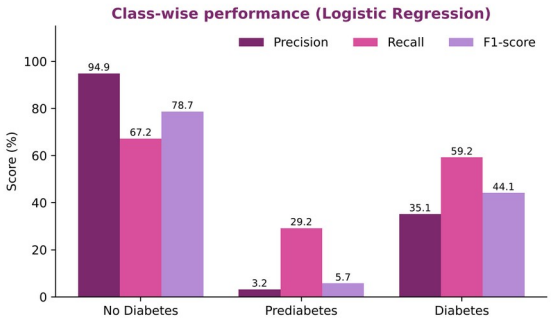


Fig. 8. Class-wise precision, recall and F1-score reported for Logistic Regression.

6.2 CONFUSION MATRIX

The confusion matrix compares the actual class with the category predicted by the model. The diagonal cells represent correct classifications, while the off-diagonal cells show records assigned to a different category. In HealthScope, the matrix makes the difficulty of separating Prediabetes from the other classes visible.

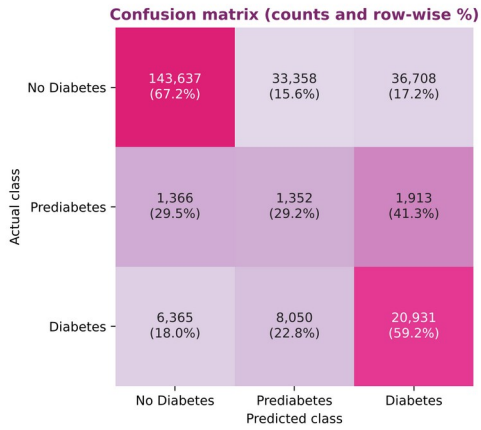


Fig. 9. Confusion matrix for the three diabetes-status categories.

6.3 INTERPRETATION OF THE RESULTS

The high precision for No Diabetes is partly understood in the context of the dataset: this category represents 84.24% of all records. The model's lower recall for this group also indicates that it does not identify every No Diabetes record. The Prediabetes category is especially difficult because it represents only 1.83% of the dataset. The results therefore point to class imbalance as an important issue for the current model.

The metrics were calculated on the complete dataset, including records used during training. They must be described as full-dataset evaluation results, not held-out test results. A separate test set and cross-validation are needed to estimate performance on unseen records more reliably.

7. EXPLORATORY DATA ANALYSIS AND FEATURE RELATIONSHIPS

Exploratory data analysis (EDA) is used in HealthScope to understand the structure of the dataset before interpreting model predictions. The dashboard presents the distribution of the target classes and charts relating diabetes status to selected health indicators. These visual summaries help users explore the data without needing to inspect every row in the CSV file.

7.1 DISTRIBUTION OF DIABETES STATUS

The target distribution shows a large majority of records in the No Diabetes category, followed by Diabetes and then Prediabetes. This imbalance is important for model development because a classifier trained on an uneven dataset may learn the majority class more effectively than the smaller classes. The distribution chart is therefore relevant not only as a descriptive statistic but also as context for the class-wise performance results.

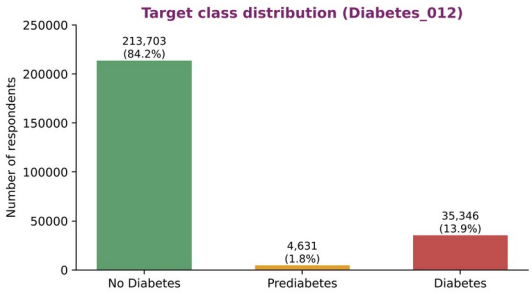


Fig. 10. Distribution of the target variable Diabetes_012.

7.2 CORRELATION WITH DIABETES STATUS

The project documentation reports positive correlations between Diabetes_012 and several indicators. The strongest listed correlations are general health (0.30), high blood pressure (0.27), BMI (0.22), difficulty walking (0.22), high cholesterol (0.21) and age (0.19). These values describe associations observed in the survey dataset. They do not establish that a particular indicator causes diabetes.

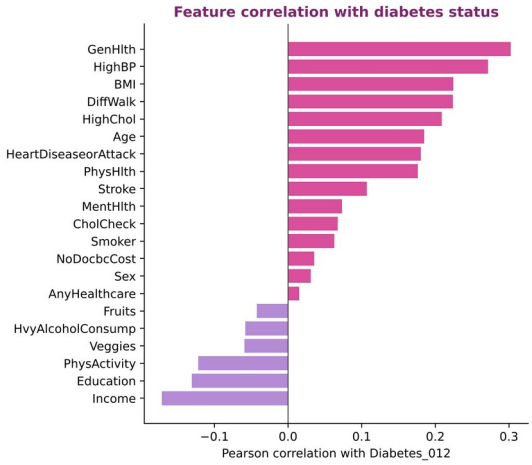


Fig. 11. Feature correlations with the Diabetes_012 target, as visualized in the project.

7.3 LOGISTIC REGRESSION COEFFICIENTS

The documentation also includes a chart of Logistic Regression coefficients for the Diabetes class. Coefficients indicate how the fitted model associates a feature with the class score, given the model's encoding and other input variables. Positive and negative coefficients show different directions of association in the fitted model. They should not be interpreted as causal effects or as standalone medical risk estimates.

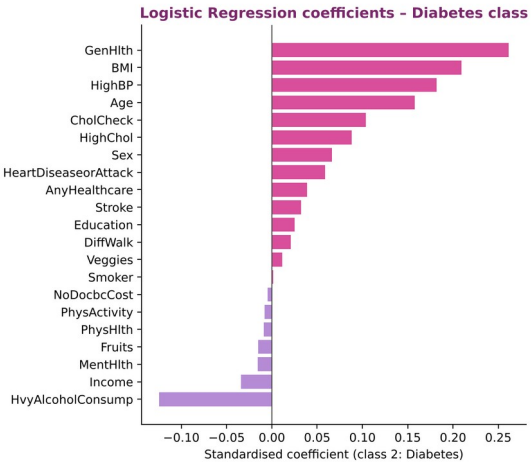


Fig. 12. Logistic Regression coefficients for the Diabetes class.

7.4 GROUP-LEVEL PATTERNS

The analytics section reports that average BMI increases from 27.74 for No Diabetes to 30.72 for Prediabetes and 31.94 for Diabetes. The proportion of records reporting high blood pressure is 37.1%, 62.9% and 75.3%, respectively. Physical activity is reported by 77.9% of the No Diabetes group, 67.8% of the Prediabetes group and 63.1% of the Diabetes group. These summaries describe differences between groups in the dataset.

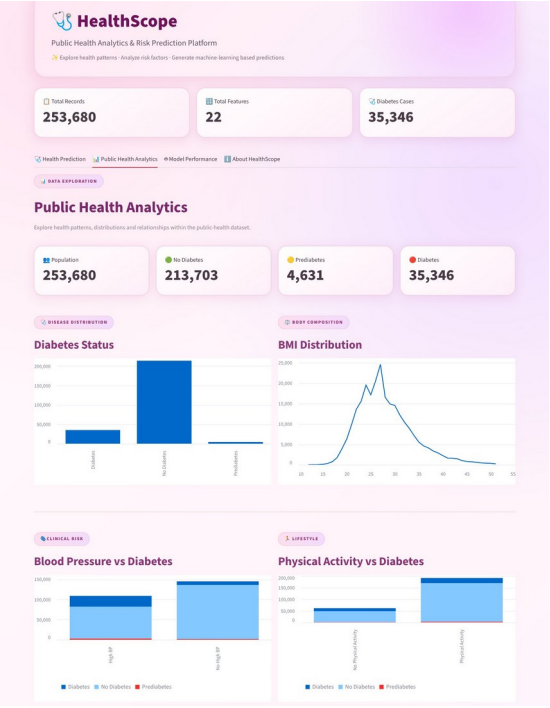


Fig. 13. Additional group-level health indicator summaries in the dashboard.

Taken together, the charts make the dataset easier to interpret and provide context for the prediction results. They also reinforce the need to avoid causal claims: survey associations may reflect several overlapping factors, and further study would be needed to explain the relationships.

8. LIMITATIONS AND FUTURE WORK

HealthScope demonstrates an end-to-end public-health analytics workflow, but the current implementation has limitations that should be considered before drawing conclusions from its output.

8.1 LIMITATIONS

- Class imbalance: Prediabetes accounts for only 1.83% of the records, and the model has low precision and recall for this category.
- Evaluation method: the reported metrics were calculated on the full dataset, including training records. They do not establish generalization to unseen data.
- Survey-based information: the dataset contains survey indicators and does not provide laboratory measurements such as blood-glucose test results.
- Single algorithm: the current project uses Logistic Regression and does not include a completed comparison with other classifiers.
- Prediction scope: the output is a model-generated category and must not be treated as a medical diagnosis.

These limitations are important because the dashboard is designed to demonstrate analysis and prediction, not to make clinical decisions. A clear explanation of the model's weaknesses is part of presenting the project responsibly.

8.2 FUTURE WORK

Several improvements can be explored in later versions. The first is to compare Logistic Regression with Decision Tree, Random Forest and K-Nearest Neighbours. Such a comparison should use the same training and test partitions and report class-wise precision, recall and F1-score so that the effect on Prediabetes can be assessed.

A second direction is to address class imbalance. Class weights or resampling methods could be tested, followed by an evaluation on a separate test set. Cross-validation would provide a more stable estimate than a single evaluation. Decision thresholds and probability outputs could also be studied, especially when the goal is to improve recognition of the minority class.

The application itself could be extended with downloadable reports, additional dashboard filters, clearer explanations of individual predictions, improved data-loading performance and cloud deployment. Any such extension should retain an explicit statement that the system is for educational and analytical use.

9. CONCLUSION

This paper presented HealthScope, a web-based platform that combines public-health data analysis with diabetes-status classification. The application uses 253,680 survey records and 21 input features to classify records into No Diabetes, Prediabetes or Diabetes. It connects a Logistic Regression model and a saved StandardScaler to a Streamlit interface, while also presenting dataset summaries, health-indicator charts and model-performance information.

The reported full-dataset evaluation achieved 65.41% accuracy and a weighted F1-score of 72.54%. Class-wise results show that the model performs much less effectively for Prediabetes than for the majority No Diabetes class. This highlights the importance of class balance and careful evaluation. HealthScope provides a useful academic demonstration of a complete analytics workflow, while further validation on unseen records is necessary before its predictions can be considered for any practical setting.

DECLARATIONS

Ethical considerations: The project uses an existing survey dataset and does not report a new clinical study or intervention involving participants. The application is not a diagnostic tool. Data availability: The dataset and implementation files are those included with the HealthScope project documentation. Conflict of interest: No competing interest is declared by the student authors. Funding: No external funding is reported.

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